Independent machine verification of the fundamental-group computation in Wuebben's exotic $S^2 \times S^2$

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Abstract

Wuebben (arXiv:2608.17267), building on Lidman–Piccirillo (arXiv:2505.14387), constructs a candidate exotic $S^2 \times S^2$ and a candidate exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ by Luttinger surgery on a non-product genus-2 surface bundle over a genus-2 surface. The critical step, historically the failure point of claimed small exotic 4-manifolds, is the triviality of eight fundamental groups obtained by filling the surgered complement. We report a verification of that step independent of the paper's coordinates, software, and author: the bundle is rebuilt as a simplicial complex from the paper's marked-fiber data alone; the surgery tori, based meridians, and framed longitudes are extracted combinatorially; and the eight filled presentations are proved trivial by complete confluent rewriting, with every computation emitting a replayable proof certificate validated by two independently written checkers. The author's committed scripts enter at exactly one point, parsed and never executed, as a cross-check of the dictionary of conventions read from the paper. The paper's framing lemma is machine-checked with its analytic normalizations rebuilt constructively and chart independence proved between the paper's displayed charts, and an exhaustive robustness scan quantifies the consequences of any framing discrepancy. The deductions downstream of simple connectivity pass a hypothesis audit. Stated carefully: the proof appears complete relative to explicitly named standard topology theorems, with no known project-specific gap. The theorem is Wuebben's; this note claims only the verification. All certificates, code, and provenance are public.

1 Introduction

Claimed constructions of small exotic 4-manifolds have repeatedly failed in one place: the fundamental-group computation. The manifolds of [1] are obtained by two Luttinger surgeries [5, 3] on Lagrangian tori in a symplectic non-product genus-2 surface bundle R over a genus-2 surface, following the framework of [2]. Exoticness rests on the surgered manifolds being simply connected: eight groups — the four sign choices of the double surgery in the paper's principal coordinate system, and four adjacent "half-drift" systems — must be trivial.

This note reports an independent verification of that step, together with a machine audit of the framing lemma that connects the group theory to the symplectic geometry and a hypothesis audit of everything downstream of simple connectivity. Independence here is concrete: nothing in the verification reuses the paper's coordinates, code, or intermediate data. The geometry was rebuilt from the paper's stated marked-fiber data alone, and the paper's displayed relations, sign tables, and Table 1 filling words were then *compared against* the rebuild rather than assumed.

Two disclosures frame everything that follows. First, the theorem is Wuebben's; this note asserts nothing about priority or authorship of the mathematics and claims only the verification. Second, the verification was carried out by AI systems (Anthropic's Claude models and OpenAI's Codex) under the author's direction, with commit-level provenance; Section 10 describes the method, and mathematical responsibility rests with the human author. No claim below depends on trusting a model: every asserted computation ships with a replayable certificate.

2 The independent model

The genus-2 fiber is realized as a regular neighborhood of the paper's five-chain of curves: five plumbed annular bands and two cone discs, with the hyperelliptic-type involution φ_0 realized simplicially. Its fixed points, the curve actions $a \leftrightarrow e$, $b \leftrightarrow d$, and the free half-rotation of the middle curve c are executable assertions about the complex, not assumptions. The bundle R is assembled from simplicial mapping cylinders over a once-punctured torus base; the two surgery tori T_α, T_β arise as induced subcomplexes. The assembled 4-complex, with f-vector [9156, 116714, 356728, 403152, 153984], passes manifold checks down to the vertex-link level.

The identification with the paper's marked bundle is by literal based monodromy action: the based mapping classes of both monodromies are certified as equalities of vertex paths, transported by proof-producing Tietze chains that replay step by step (17,839 elementary steps for the primary model; 34,735 for an independently built second triangulation). The identification is completed by an explicit marked mapping-cylinder clutching over the base graph: the relative monodromy comparisons are exact on the marked collars, so the identification is a constructed fiberwise homeomorphism into the paper's bundle rather than an appeal to the classification of surface bundles. Everything the surgery argument needs is transported through that map into the paper's already-smooth manifold, where the smooth and symplectic arguments run; no smoothing of the triangulation is ever chosen.

3 Peripheral data

Based meridians and product-framing longitudes for both tori are traced as literal simplicial loops carrying the paper's own basing whiskers. The paper's sign-correction package (its §7.4 fix) is reproduced independently. A combinatorial sweep of the relevant transport square resolves the basing double-count question in the direction the paper's corrected convention requires: the directly computed boundary longitude already contains the geometric basing effect, and must not be multiplied by an extra meridian.

Both surgery tori are certified locally flat at all 1,776 of their simplex links — 296 vertex, 888 edge, and 592 triangle links, each classified against the correct unknotted model — and the boundary carrying the peripheral curves is an explicit half-weight normal block model with a global PL identification of the computed frontier: every extracted dual meridian is checked to be a literal normal-circle fiber, with no regular-neighborhood theorem invoked.

4 The eight filled groups

Filling the triangulation-derived complement presentation (4 generators, 95 relators) with the certified coherent peripheral pairs yields, for all four $n = 0$ sign pairs and the four adjacent half-drift systems, complete confluent rewriting systems in which every generator reduces to the identity. Knuth–Bendix completion (KBMAG [11], run under GAP [12]) is used only as an *untrusted certificate generator*: the exported derivation DAGs are validated by a small independent checker that re-verifies every input-relator axiom, inverse axiom, critical overlap, rewrite trace, and final generator-to-identity rule, and binds the input presentation by SHA-256. The checker neither imports nor runs KBMAG. A second implementation, written independently in Ruby, re-verifies all 14,115 retained derivation records against the same hash-bound DAGs and accepts, importing neither the Python checker nor any project group code.

system	signs	records	system	signs	records
n=0	$(-,-)$	1123	n=1	$(-,-)$	1709
n=0	$(-,+)$	1075	n=1	$(-,+)$	3408
n=0	$(+,-)$	804	n=1	$(+,-)$	1975
n=0	$(+,+)$	1504	n=1	$(+,+)$	2517

Derivation-DAG certificates for the eight filled groups; each verifies TRIVIAL from a fresh clone of the repository.

All artifacts — the eight certificates, the exported rewriting systems, and a 23-hash manifest — replay from a clean checkout, and the stored rewriting systems reload and re-verify under GAP/KBMAG independently of the certificate path.

5 Redundancy against translation error

Because the machine certifies properties of a model, the model itself is the residual risk. Five structurally different implementations attack it: an independent peripheral extractor built only from the bundle; an alternative bundle triangulation with a different subdivision; a second based-monodromy certificate on that alternative; a simplex-level verifier of all 128 PL flip traces; and a second marked-fiber realization, built directly from the abstract ribbon graph with a different vertex count (58 versus 86) and matched equivariantly to the primary model. A single translation mistake would need to survive all of them. What redundancy cannot exclude — a shared misreading of the paper's conventions — is attacked separately by exact reproduction of the paper's displayed relations and sign tables, re-audited against the typeset PDF rather than any text extraction. The translation layer is itself a bound artifact: an interpretation dictionary in which every convention read from the paper carries the certified computation that would fail under a rival reading, with the central entries cross-checked — parsed, never executed — against the author's own committed scripts, which define the same monodromy action, corrected relations, and sign conjugators. The three entries that reading could pin only by transcription — the ordered beta twists, the five-chain ribbon labels, and the local twist signs — were then attacked at the raw-paper level: a quarantined extractor reads only the paper's plain text, reconstructs the five-chain, involution, ordered twists, surgery tori, whiskers, and direction words, and a separate process compares its frozen output with the independently built model certificates, with reversed-order and reversed-sign mutations distinguished. No discrepancy was found: the previously identified transcription residuals are closed relative to the paper's stated marked data. The source itself was then audited directly rather than accepted through the paper's restatement: a checker reading only the hash-pinned `main.tex` and original vector Figure 1 of the immutable Lidman–Piccirillo v1 arXiv source [2], importing no project module, confirms the ordered five-chain a, b, c, d, e, the chain-reversing involution with c fixed with orientation, the single consecutive crossings with all required disjointness, and the two fixed points, in both the prose and the vector layers. The one interpretation remaining at that boundary is the ordinary convention that dashed arcs in a handle picture depict hidden projection rather than extra crossings. The project-specific content of the paper's Lemma 7.1 is certified in the same sense: all 36 candidate equivariant ribbon rotations are enumerated, exactly four survive and all normalize to one marked ribbon with two invariant free-half-turn boundary disks, so the lemma's finite ribbon classification and its extension hypotheses are machine-checked, with the Kerékjártó periodic-disk involution theorem [4] the remaining named input now that the source marked data are audited. Each of these checks freezes its output as a certificate and logs its run transcript; all of it replays directly from the repository.

6 The framing lemma

Luttinger surgery is performed with respect to the Lagrangian framing; the paper's Lemma 8.2 identifies it with the fibered framing carried by the combinatorial longitudes of Section 4. This lemma carries an exact logical load: if the identification failed by j meridians, the certificates of Section 4 would remain correct for the product-framed presentations they literally describe, but would no longer identify those presentations with the paper's surgeries, and simple connectivity, the homeomorphism conclusions, and the exoticness argument would all become unproved.

The lemma's displayed Weinstein-chart, seam, and constant-momentum algebra are machine-checked symbolically, and its inline analytic normalizations have been rebuilt constructively rather than cited. The relative Moser flow is produced as the explicit monotone inverse of the cumulative coordinate $H_s(\theta, t) = t + s\int_0^t (f - 1)$, whose positivity yields a uniform invariant collar; differentiating the defining identity produces the Moser equation and the pullback formula, so no ODE existence theorem is invoked. The equivariant step defines the Moser field directly on the double cover, with certified projection, deck invariance, and the factor-two normalization, so no covering-lift theorem is invoked; the covering hypotheses themselves — free involution, genuine 2:1 simplicial covering, deck group exactly $\{\mathrm{id}, \varphi_0\}$, core winding number two — are certified on the actual simplicial collar.

Chart independence is argued twice within the project: a fiber-dilation isotopy takes any chart-transition germ to the identity relative to the zero section, and a construction-side proof shows the paper's charts are forced by pairing the Liouville primitive $t\,d\theta_1 + Ku\,d\theta_2$ with the well-defined angle frames, with every integral change of angle basis preserving constant momentum and the one symplectic move that could introduce meridian components — the closed-1-form shift — excluded by the zero-section condition. [3] (§2.1 and Proposition 2.2) stands as corroboration. After this program, no standard input to the lemma remains cited rather than mechanized: the only form of Weinstein germ uniqueness the argument uses is the statement proved above, the independence of the constant-momentum push-off between the paper's displayed charts.

7 The framing-robustness scan

Independently of the lemma, we quantify what a framing discrepancy would do. For each discrepancy pair (j_a, j_b), the complement is refilled with the meridian-shifted longitudes $\lambda\mu^j$, imposed with each torus's own certified based meridian, and the identical certification pipeline is run: Tietze collapse, Knuth–Bendix completion, and — for resistant cases — abelianization, quotient search onto A_5, $\mathrm{PSL}(2,7)$, A_6, $\mathrm{PSL}(2,8)$, $\mathrm{PSL}(2,11)$, $\mathrm{PSL}(2,13)$, low-index subgroups to index 8, and coset enumeration (ACE [13]) to a ceiling of 12,499,998 cosets. The $(0,0)$ controls reproduce the certified filling relators word for word.

	(j_a, j_b):	0, 0	1, 0	-1, 0	2, 0	-2, 0	0, 1	0, -1	0, 2	0, -2	1, 1	1, -1	-1, 1	-1, -1
n=0	$(+,+)$	K	·	T	·	K	K	T	K	K	·	T	T	T
	$(+,-)$	K	K	T	·	K	T	K	K	K	T	·	T	T
	$(-,+)$	K	T	K	K	·	K	T	K	K	T	T	K	T
	$(-,-)$	K	T	K	K	K	T	K	K	K	T	T	T	·
n=1	$(+,+)$	—	·	T	·	K	K	T	K	K	·	T	T	T
	$(+,-)$	—	·	T	·	K	T	K	K	K	T	·	T	T
	$(-,+)$	—	T	·	K	·	K	T	K	K	T	T	·	T
	$(-,-)$	—	T	K	K	·	T	K	K	K	T	T	T	·

T = trivial by proof-producing Tietze collapse; K = trivial by complete confluent rewriting; · = undecided (perfect, witness-free, enumeration-resistant); — = the n=1 controls are the certified half-drift fillings of Section 4.

Of 100 cases, 82 are certified trivial (40 by Tietze collapse, 42 by confluent rewriting), 18 are undecided, and *none is nontrivial.* Every undecided group is perfect, admits no epimorphism onto the six smallest nonabelian simple groups, has no proper subgroup of index at most 8, and resists coset enumeration to the stated ceiling. The scan's structural finding: all 32 shifts touching only T_β are certified trivial in both families — a framing error on the second torus alone provably cannot change the simple-connectivity conclusion — and every undecided case involves a shift on T_α. We emphasize that these are counterfactual groups: the paper's own fillings are the certified $(0,0)$ column. The 18 undecided counterfactual groups are not evidence of a framing error, and they have no bearing on the $j_a = j_b = 0$ proof unless the independent framing argument of Section 6 first fails.

8 The downstream audit

Conditional on the certified simple connectivity, the deductions of the paper's main theorem pass a hypothesis audit: the covering exact sequence and Hambleton–Kreck classification data [7, 8], as summarized in [9]; the even and odd rank-two intersection-form computations feeding Freedman's classification [6]; minimality and symplectic Kodaira dimension by two independent routes; the symplectic Thom genus bound behind the no-torus lemma; the lifted-torus and Rokhlin/Arf steps of the slicing argument; and the reuse of the Lidman–Piccirillo relative Floer argument [2], including why the stronger π_1 statement is the one the regluing needs. The elementary lattice, Euler characteristic, cover-genus, and adjunction computations are executable.

9 Trust boundary

The named standard inputs, applied with checked hypotheses and cited proofs: the elementary ribbon-thickening and bistellar-trace interpretations that read the paper's figures into complexes; the classification of surfaces; the collapsible-3-ball and cyclic-knot-unknot criteria behind the local-flatness certificates; simplicial fundamental-group presentation and Tietze theory; the Kerékjártó periodic-disk involution theorem [4] behind Lemma 7.1's disk extension; Freedman's simply connected classification [6] with the Hambleton–Kreck finite-cyclic classification [7, 8]; symplectic Kodaira dimension and minimality, via the audited Luttinger-surgery constructions [3, 10]; the symplectic Thom theorem; the Rokhlin/Arf input of the slicing argument; and the Ozsváth–Szabó / Lidman–Piccirillo Floer machinery [2]. This note does not reprove them. The ribbon-interpretation input is narrower than the phrase suggests: the redundancy section's classification certifies the implication from the paper's stated marked five-chain data to the equivariant normal form, and the stated data themselves are audited in the original Lidman–Piccirillo source, so what this boundary retains is the thickening of that data into complexes together with the ordinary dashed-arc hidden-projection convention of the audited figure.

Formerly listed inputs that are no longer dependencies: surface-bundle classification, replaced by the explicit clutching; four-dimensional smoothing and intersection naturality, since the smooth arguments run entirely in the paper's already-smooth manifold; derived regular-neighborhood theory, since the tubular boundary is constructed; and Weinstein germ uniqueness, since chart independence is proved between the paper's verified charts (Section 6), with [3] as corroboration. The accurate one-sentence status is: *the proof appears complete relative to explicitly named standard topology theorems, with no known project-specific gap* — a statement about verification depth, offered to make expert review cheap, not to replace it.

10 Provenance and method

The verification was AI-assisted throughout: Anthropic's Claude models (Fable 5 and Opus 5) and OpenAI's Codex, working under the author's direction, with a commit-level provenance ledger recording which model produced which artifact. Cross-model review was adversarial by design, and several decisive corrections in the record came from one model refuting the other's assessment; the ledger preserves those corrections rather than rewriting them. Mathematical responsibility rests with the human author. The design principle throughout is that certificates are checked, not trusted: search and completion tools are treated as untrusted generators, and every claim rests on a replayable artifact validated by small independent code.

11 Data availability

All code, certificates, run transcripts, referee packets, and the provenance ledger are at `https://github.com/VentiMath/exotic-s2xs2-verification`, including clean-checkout replay instructions. The paper's own code is at `https://github.com/bwuebben/exotic-s2xs2`; nothing in the verification derives from it except where explicitly cited.

References

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